DO YOU KNOW?

Magnified images of some objects:



Coloured scanning electron micrograph of a common housefly (Muscadomestica).



Needle and thread

Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626\times10^{-34}}{7.63\times10^{-25}} = 0.868\times10^{-9}\,\text{m}$$
$$= 8.68\ \text{Å}$$

ii) Momentum of the bullet is

$$p = mv = 0.050\times200 = 10\ \text{kgms}^{-1}$$

Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626\times10^{-34}}{10} = 6.626\times10^{-35}\,\text{m}$$

iii) Momentum of the car is

$$p = mv = 4000\times50 = 2\times10^{5}\,\text{kgms}^{-1}$$

Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626\times10^{-34}}{2\times10^{5}} = 3.313\times10^{-39}\,\text{m}$$

From these calculations, we notice that electron has significant value of de Broglie wavelength ($\approx10^{-9}$m which can be measured from diffraction studies) but moving bullet and car have negligibly small de Broglie wavelengths associated with them ($\approx10^{-33}$m and 10^{-39}m respectively, which are not measurable by any experiment). This implies that the wave nature of matter is important at the atomic level but it is not really relevant at the macroscopic level.

EXAMPLE 8.7

Find the de Broglie wavelength associated with an alpha particle which is accelerated through a potential difference of 400 V. Given that the mass of the proton is 1.67×10^{-27} kg.

Solution

An alpha particle contains 2 protons and 2 neutrons. Therefore, the mass M of the alpha particle is 4 times that of a proton (m_p) (or a neutron) and its charge q is twice that of a proton ($+e$).

The de Broglie wavelength associated with it is

$$\lambda = \frac{h}{\sqrt{2MqV}} = \frac{h}{\sqrt{2\times\left(4m_p\right)\times\left(2e\right)\times V}}$$
$$= \frac{6.626\times10^{-34}}{\sqrt{2\times4\times1.67\times10^{-27}\times2\times1.6\times10^{-19}\times400}}$$
$$= \frac{6.626\times10^{-34}}{4\times20\times10^{-23}\sqrt{1.67\times1.6}} = 0.00507\ \text{Å}$$

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